

14.1 CHARACTERISTICS OF SOUND

LO 2

Sound is characterised by the following parameters:

1. its intensity in W/m^2 or its pressure in μbar ,
2. its frequency in cycles per second or Hz,
3. the nature of noise which may be either continuous or impulsive,
4. its loudness or loudness level, and
5. its *noise rating* which is also called *annoyance* rating based on the sound pressure level and frequency.

Characteristics (1), (2) and (3) are measured objectively by the instruments and do not depend on the subjective assessment of the observer; whereas (4) and (5) depend on the subjective assessment of the quality of sound by the human observer which may be based on measured results, set of standard tables, graphs, etc.

14.2 SOUND PRESSURE, POWER AND INTENSITY LEVELS

LO 3

14.2.1 Sound Pressure Level (SPL)

The audible range of human hearing mechanism usually measured at 1 kHz, extends from $0.0002 \mu\text{bar}$ at the *threshold of hearing* to 1 mbar at the *threshold of pain*. This represents an increase of 5×10^6 . Because of this very large range, it is more convenient to express the magnitude of sound pressure in logarithmic form, in terms of decibels.

Thus, sound pressure level is often abbreviated as SPL or L_p and is defined in decibels as

$$\begin{aligned}\text{SPL} &= 10 \log_{10} (p^2)/(p_{\text{ref}}^2) \\ &= 20 \log_{10} (p)/(p_{\text{ref}}) \text{ dB}\end{aligned}\quad (14.1)$$

where p is the root mean square (rms) sound pressure and p_{ref} the rms reference pressure ($p_{\text{ref}} = 0.0002 \mu\text{bar} = 0.00002 \text{ N/m}^2 = 0.0002 \text{ dyn/cm}^2$). A reference pressure of $0.0002 \mu\text{bar}$ has been chosen as the smallest possible sound level at 1 kHz that a young adult can just hear (in quiet surroundings). In other words, the *threshold of hearing* is taken as the reference pressure of SPL measurements. Hence, Eq. (14.1) can be written as

$$\text{Sound pressure level (SPL)} = 20 \log \frac{p}{0.00002} \text{ dB} \quad (14.2)$$

where p is in N/m^2 .

The advantage of using logarithmic scale in defining the SPL is that it permits a large range of pressures to be described with convenience. Further, it also represents the non-linear behaviour of the ear more convincingly.

SPL is measured using a sound level meter. Figure 14.1 gives pictorially the typical values of the order of Sound Pressure Levels (SPLs) of some common sounds.



Problem 14.1

If the pressure of sound measured at 1 m from an unmuffled motor cycle was found to be 12.0 N/m^2 , determine the corresponding sound pressure level in decibels, p_{ref} is $2 \times 10^{-5} \text{ N/m}^2$.

Solution Using Eq. (14.2) we get,

$$\text{SPL} = 20 \log \frac{12.0}{0.00002} = 115.56 \text{ dB}$$

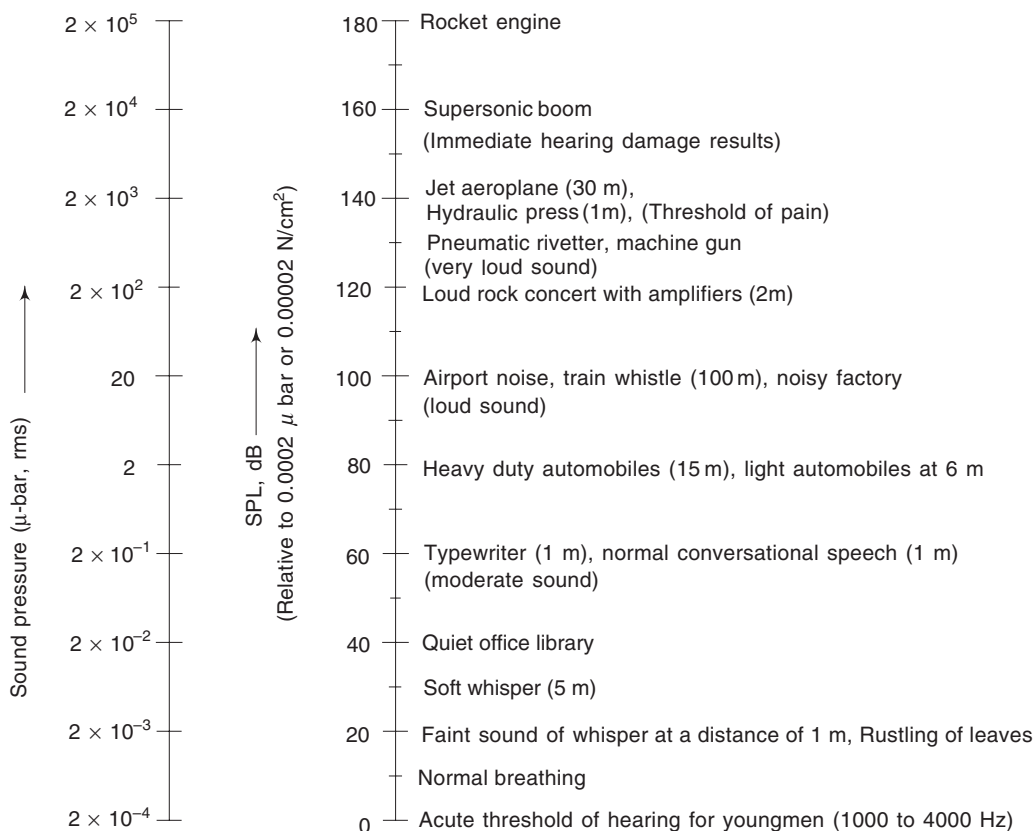


Fig. 14.1 Typical values of sound pressure levels of common sounds with reference sound pressure of $0.002 \mu\text{bar}$, the threshold of hearing at 1 kHz

14.2.2 Sound Power Level (PWL)

Sound power is the total sound energy radiated by a sound source per unit time. The power in the sound emitted from a source can have an extremely large range. The power can be as low as 10^{-9} W at a faint whisper or 10^{-5} W at normal conversational speech. Further, it could be as large as 50 kW for a jet liner or 10 MW for a saturn rocket. Because of this wide range of values of power in the various power sources, the sound power level is conveniently defined in the units of decibels. Sound power level is often abbreviated as PWL or L_w and is defined as:

$$\text{PWL} = 10 \log_{10} \frac{W}{W_{\text{ref}}} \text{ dB} \quad (14.3)$$

where W is acoustic power of the source and W_{ref} is the reference acoustic power ($W_{\text{ref}} = 10^{-12} \text{ W}$). Since the power ratio 10^{-12} means -120 dB , the above equation may be written as:

$$\text{PWL} = (10 \log_{10} W + 120) \text{ dB} \quad (14.4)$$

where W is in watts.

Figure 14.2 gives the typical values of the order of sound power levels of some common sounds.

It is interesting to note that small sound power levels can have a significant effect on the ear. It is estimated that the acoustic power emitted by the roars of the football crowd during a match is just enough

to heat a cup of coffee! Further, the power of a full symphony orchestra is only one acoustic watt or so!

Sound pressure level (SPL) of a machine, that is radiating noise, depends on the machine as well as on the surroundings. The same machine may show different SPLs when placed in different surroundings. It is for this reason that sound power level (PWL) has been defined. PWL cannot be directly defined but is found by the measurement of sound pressure levels or sound intensity levels.

When there are more than one acoustic power sources, acoustic power W of all sources can be added and the combined PWL can be found using Eq. (14.3).

14.2.3 Sound Intensity Level (L_I)

Sound intensity I at a location in a sound field in a specified direction, is the average sound power passing through a unit area for which the level is being specified. Its unit is W/m^2 .

Sound intensity level L_I is defined:

$$L_I = 10 \log_{10} \frac{I}{I_{\text{ref}}} \text{ dB} \quad (14.5)$$

The value of I_{ref} is taken as 10^{-12} W/m^2 in Eq. (14.5).

14.2.4 Relation between Sound Pressure Level (SPL) and Sound Intensity Level (L_I)

The intensity of sound I for a progressive sound wave in air along the direction of wave propagation is given by

$$I = \frac{p_{\text{rms}}^2}{\rho v} \quad (14.6)$$

where p_{rms} is the sound pressure in N/m^2 , ρ is the density of medium in kg/m^3 , and v is the velocity of sound in the medium in m/s .

The denominator of Eq. (14.6) i.e., product of (ρv) is defined as the characteristic impedance of the medium.

At a temperature of 20°C and with standard atmospheric conditions, the characteristic impedance of air is given by

$$\begin{aligned} (\rho v) &= (1.21 \text{ kg/m}^3) (343 \text{ m/s}) \\ &= 415 \text{ kg/m}^2 \end{aligned} \quad (14.7)$$

Substituting the value of (ρv) from Eq. (14.7) in Eq. (14.6) we get,

$$I = \frac{p_{\text{rms}}^2}{415} \quad (14.8)$$

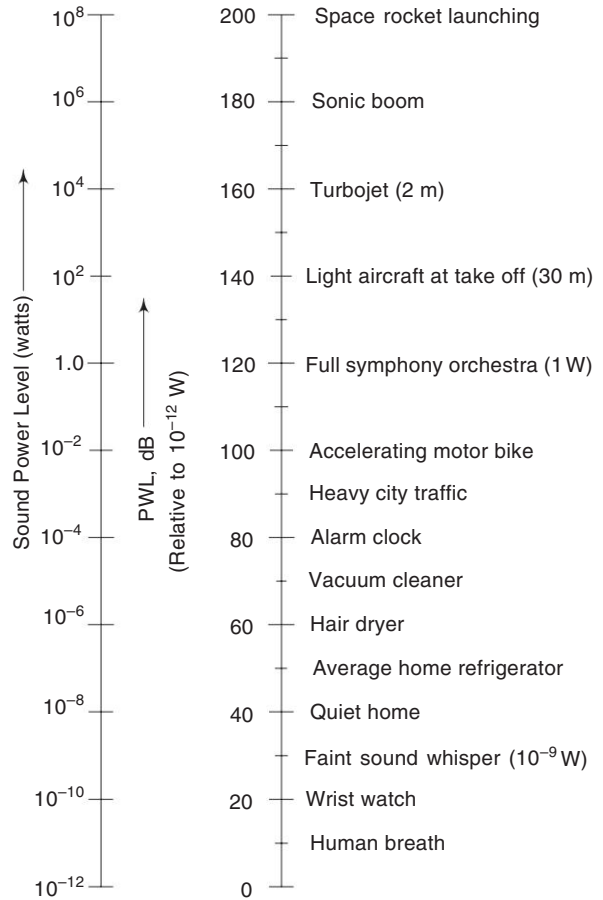


Fig. 14.2 Typical Values of Sound Power Levels (PWLs) of common sounds with reference to power of 10^{-12} W

Multiplying the numerator and denominator of Eq. (14.5) with $(p_{\text{ref}}^2)_{\text{rms}}$ and substituting the value of I from Eq. (14.8), we get,

$$\begin{aligned}
 \text{Sound Intensity Level } L_I &= 10 \log_{10} \left[\frac{p_{\text{rms}}^2}{(p_{\text{ref}}^2)} \right] \left[\frac{(p_{\text{ref}}^2)_{\text{rms}}}{415 I_{\text{ref}}} \right] \\
 &= 20 \log_{10} \left[\frac{p_{\text{rms}}}{(p_{\text{ref}})_{\text{rms}}} \right] + 10 \log_{10} \left[\frac{(p_{\text{ref}})_{\text{rms}}^2}{415 I_{\text{ref}}} \right] \\
 &= \text{SPL} + 10 \log_{10} \left[\frac{(p_{\text{ref}})_{\text{rms}}^2}{415 I_{\text{ref}}} \right] \quad (14.9)
 \end{aligned}$$

Substituting the value of $p_{\text{ref}} = 2 \times 10^{-5} \text{ N/m}^2$ and $I_{\text{ref}} = 10^{-12} \text{ W/m}^2$ in Eq. (14.9), we get,

$$\begin{aligned}
 L_I &= \text{SPL} + 10 \log_{10} \left[\frac{(2 \times 10^{-5})^2}{(415)(10^{-12})} \right] \\
 &= \text{SPL} + 10 \log_{10} \left(\frac{400}{415} \right) \\
 &= \text{SPL} - 0.16 \quad (14.10)
 \end{aligned}$$

Therefore, for all practical purposes, the *Sound Pressure Level (SPL)* is numerically taken to be equal to *Sound Intensity Level L_I* .

14.2.5 Addition of Sound Pressure Levels

If there are two or more sources radiating noise, the total sound pressure level may be determined from the individual sound pressure levels of the sources. It may be done by addition based on energy or power levels which are dependent on mean square pressures. Thus, the SPL of each based may be converted

from number of decibels to $\left(\frac{p^2}{p_{\text{ref}}^2} \right)$ values, by dividing the decibels by 10 and calculating antilogs. After

adding the relative (p^2/p_{ref}^2) values of various sources, the conversion to dB levels may be made for the combined sound pressure levels, using Eq. (14.1).

Let us consider two such sound pressure levels of X dB and Y dB.

$$\therefore \text{Resultant SPL} = 10 \log \left[\text{antilog} \left(\frac{X}{10} \right) + \text{antilog} \left(\frac{Y}{10} \right) \right] \quad (14.11)$$

Similarly, if there are several sources, with SPL's $X, Y, Z, \dots$, etc. then

$$\text{Total SPL} = 10 \log \left[\text{antilog} \left(\frac{X}{10} \right) + \text{antilog} \left(\frac{Y}{10} \right) + \text{antilog} \left(\frac{Z}{10} \right) + \dots \right] \quad (14.12)$$

Now if in Eq. (14.11) $X = Y$, i.e. there are two sound sources of equal SPL, then,

$$\begin{aligned}
 \text{Total SPL} &= 10 \log \left\{ 2 \text{antilog} \left(\frac{X}{10} \right) \right\} \\
 &= 10 \left\{ 0.301 + \left(\frac{X}{10} \right) \right\} \\
 &\approx X + 3 \text{ dB} \quad (14.13)
 \end{aligned}$$

indicating sound pressure level at different frequencies. For example, if the magnitude of sound pressure level at 1000 Hz is 40 dB (loudness level equal to 40 phons), then for an equal loudness level at 100 Hz, the average person will require an SPL of 52 dB.

14.4 TYPICAL SOUND MEASURING SYSTEMS

LO 5

14.4.1 Sound Level Meter

Sound pressure levels are measured by sound level meters, which convert acoustic pressure into a directly proportional voltage indicated on a meter. The meter is positioned at the desired location with no obstruction from the sound source and the reading is taken. The block diagram of the instrument is shown in Fig. 14.4. This includes a microphone, an electronic amplifier with frequency weighting network and a meter or recorder calibrated in decibels.

A rectifier circuit is incorporated to produce a signal proportional to root mean square value. The input signal generated by the sound pressure alternates from positive to negative levels. This signal is squared, which has the effect of producing all positive signals varying in amplitude. By taking the square root of this signal, it is possible to obtain the output of the meter in the form of averaged root mean square value of the signal.

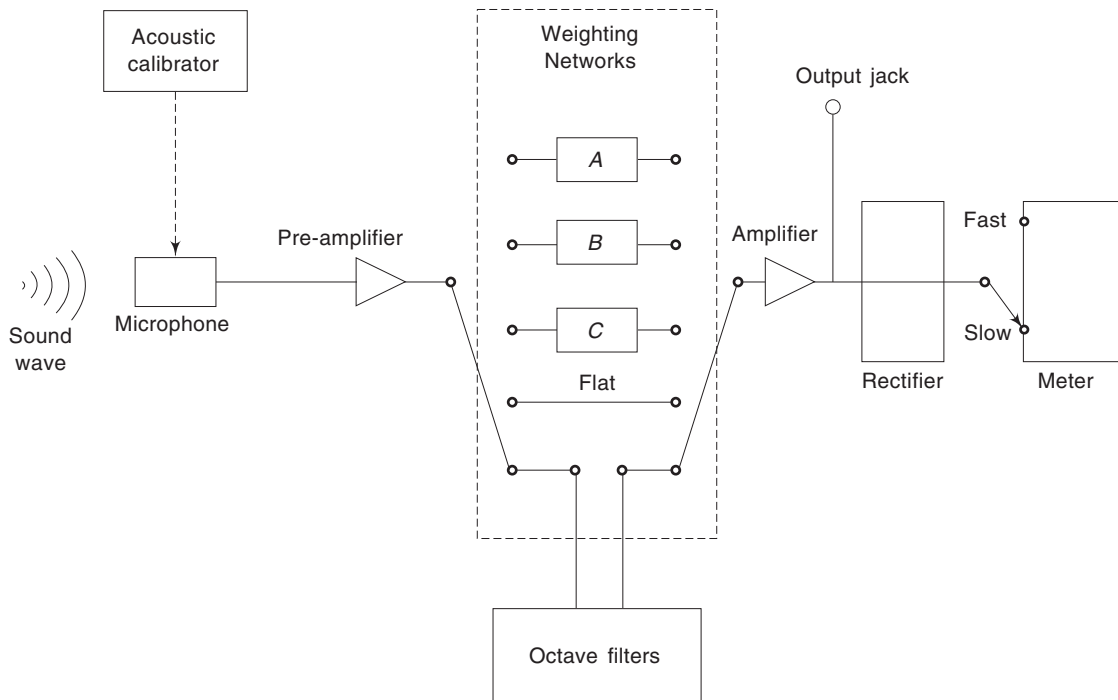


Fig. 14.4 A block diagram of a typical Sound Level Meter (SLM)

where ω is the angular frequency, $I_{\text{mag}} (G_{12})$ the imaginary part of cross-spectrum between the two microphone signals, ρ the medium density and Δr in the spacing between the microphones. A computer with software is normally used for calculations and plotting. The measurements are carried out with distance r from the noise radiating surface being 40 – 80 mm. For measurements, the noise radiating surfaces are divided into suitably spaced grades and intensity measurements are done at each point of the grid by direct use of Eq. (14.6), from which intensity maps can be obtained indicating the regions of high sound radiation. In another method called sweeping or scanning method, the probe is slowly swept (usually, 0.2 m/s) over the surface while the analyser is averaging the measurements.



Problem 14.6

In a certain sound level meter, the output voltage was directly proportional to the sound pressure generated by a sound source. The instrument was calibrated with a sound pressure excitation of 0.1 N/m^2 and the meter scale was set at -60 dB referenced to 1 V .

For a given noise source, the output voltage developed by the instrument is 6 mV . Determine the SPL of the noise source.

Solution The output voltage of the sound level meter (SLM) for a reference excitation of 1 V is given by

$$\begin{aligned} -60 \text{ dB} &= 20 \log E/E_{\text{ref}} \\ &= 20 \log E/1.0 \quad (\because E_{\text{ref}} = 1 \text{ V}) \end{aligned}$$

or

$$E = 10^{-3} \text{ V}$$

Now, 6 mV output voltage of SLM $= 6 \times 10^{-3}/10^{-3} = 0.6 \text{ N/m}^2$

The sound pressure level corresponding to the acoustic pressure of 6 dyn/cm^2 is given by:

$$\begin{aligned} \text{SPL} &= 20 \log \frac{p}{p_{\text{ref}}} \\ &= 20 \log \frac{0.6}{2 \times 10^{-5}} \quad (\because p_{\text{ref}} = 2 \times 10^{-5} \text{ N/m}^2) \\ &= 89.54 \text{ dB} \end{aligned}$$



Problem 14.7

The noise signal of a certain machinery in a workshop was processed in an audio-frequency analyser which was equipped with the standard octave filters. The results obtained were as follows:

Centre frequency (in Hz)	63	125	250	500	1000	2000	4000	8000
SPL (in dB)	84	85	84	81	72	68	66	52

Determine the overall sound pressure level in dB.

Solution Using Eq. (14.8) we add the sound pressure levels of each centre frequency in the octave band and get,

$$\begin{aligned} \text{Overall SPL} &= 10 \log [\text{antilog}(8.4) + \text{antilog}(8.5) + \text{antilog}(8.4) + \text{antilog}(8.1) + \text{antilog}(7.2) \\ &\quad + \text{antilog}(6.8) + \text{antilog}(6.6) + \text{antilog}(5.2)] = 89.87 \text{ dB} \end{aligned}$$

14.5 MICROPHONES

LO 6

Microphones are transducers to convert sound pressure variations into the analogous electrical signals. The main factors in the selection of a microphone as sensitivity, frequency response, dynamic range and linearity. Microphones generally employ a thin diaphragm to convert pressure into motion. The motion is then converted into a suitable electrical output by employing a secondary transducer which may be either of the following:

1. Condenser or capacitor type microphone,
2. Electret microphone,
3. Piezo-electric microphone,
4. Electrodynamic type microphone.

14.5.1 Condenser or Capacitor Type Microphone

The capacitor or condenser type microphone is most widely used for precision measurements because of good frequency response, excellent linearity and large dynamic pressure range. A typical construction of this type microphone is shown in Fig. 14.7. The microphone cartridge consists of thin metallic diaphragm in close proximity to a rigid (stationary) back plate. The diaphragm and the back plate form the plates of the capacitor. The movement of the diaphragm caused by the impingement of sound pressure results in an output voltage given by

$$E \propto Qd \quad (14.21)$$

where E = output voltage of the capacitor,

Q = charge provided by the polarising voltage (nearly constant) and

d = separation of the plates.

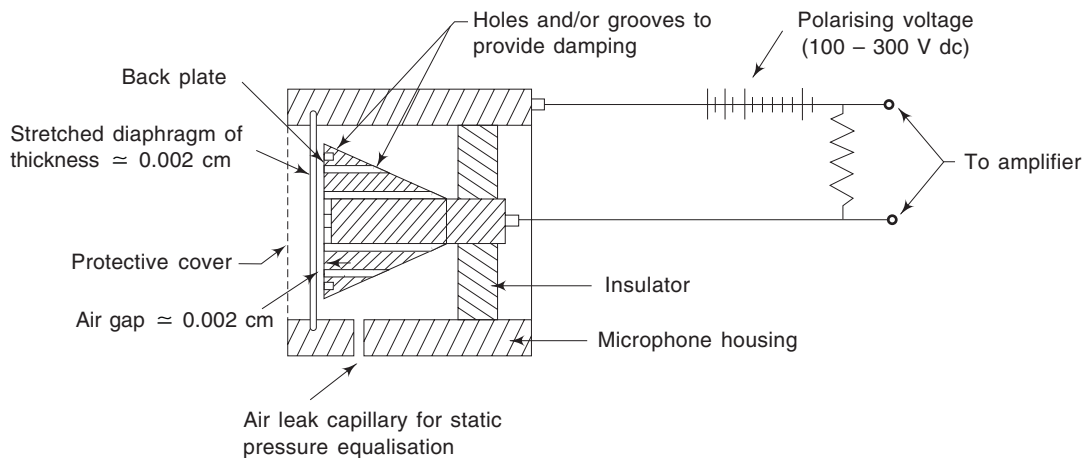


Fig. 14.7 A Schematic diagram of a condenser type microphone

The microphone is provided with a capillary air-leak for equalisation of pressure on both sides of the diaphragm to prevent the diaphragm from bursting. Further, the back plate of the condenser microphone is provided with damping holes. The motion of the diaphragm causes air flow through the holes which results in energy dissipation because of fluid friction. This damping effect is utilised in the control of resonant peaks of diaphragm response.

These microphones have good frequency response even at high frequencies and also have good acoustic sensitivity. However, they are susceptible to errors caused by high humidity, which produces excessive background noise due to the leakage of the charge.

14.5.2 Electret Microphone

Electret condenser microphone is an improved version of ordinary condenser microphone. In the condenser microphone we need to supply a dc polarising voltage, whereas in electret microphone, the polarized